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Mechanistic Interpretability of Bearing Remaining Useful Life Models via Top-k Sparse Autoencoders: Bridging Deep Learning and Vibration Theory

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Abstract

Deep learning achieves impressive accuracy in remaining useful life (RUL) prediction for rolling bearings, yet its black-box nature blocks industrial adoption: maintenance engineers need evidence that predictions are grounded in known degradation physics rather than statistical artefacts. This work (i) designs a falsifiable, statistically grounded procedure that maps a trained model's latent features to the four bearing characteristic frequencies (BPFO, BPFI, BSF, FTF) without fault labels; (ii) verifies with negative controls and bootstrap inference that the resulting latent–physics correspondence emerges only from learned representations and is robust to multiple-testing and threshold choice; and (iii) shows that the procedure generalises across heterogeneous architectures and benchmarks at negligible computational overhead. We adapt mechanistic interpretability—pioneered for large language models—to bearing prognostics by training a Top-k Sparse Autoencoder (SAE) post-hoc on RUL-model hidden states and correlating its sparse features with the Hilbert envelope spectrum at each characteristic frequency. The methodology is validated on three architectures (Mamba-xLSTM, N-BEATS-xLSTM, SparseGate-TCN) and four benchmarks (PHM2012, XJTU-SY, IMS, CWRU). Bootstrap 95 % confidence intervals, permutation tests, and three-seed (42/43/44) pooled negative controls (untrained backbone and Gaussian-noise input) confirm that the correspondence is statistically significant ($p < 0.001$) and emerges only from trained models. On PHM2012, BPFI (2.3 %) and BPFO (2.0 %) dominate, matching race spalling; on XJTU-SY, BPFO (2.2 %) leads with a residual BSF contribution (0.3 %), matching documented outer-race spalling on LDK UER204 bearings; on IMS, BPFI (1.76 %) and BSF (0.49 %) are the only significant frequencies, matching the Rexnord ZA-2115 inner-race-and-ball profile. CWRU produces a suggestive BPFI-dominant profile (5.08 %), but its small recording pool (10 fault conditions) keeps p > 0.05 and motivates excluding CWRU from the formal falsification claim. The dominant BPFx ordering is preserved across the three backbones (4–8× magnitude variation) and the procedure adds negligible overhead (≈50 minutes of SAE training per model–dataset pair on one NVIDIA A40 GPU), offering a principled bridge between data-driven RUL prediction and classical vibration theory for trustworthy predictive maintenance.

Keywords: *bearing characteristic frequency; deep learning; envelope analysis; mechanistic interpretability; predictive maintenance; remaining useful life; rolling element bearing; sparse autoencoder*

Introduction

Rolling element bearings are the most widely used and the most failure-prone components in industrial rotating machinery, including electric motors, pumps, compressors, and gearboxes that drive modern manufacturing lines and energy systems (Lei et al., 2018; Nectoux et al., 2012). Empirical surveys consistently attribute 40–50 % of all rotating-machine failures to bearing degradation, with cascading consequences for production uptime, supply-chain resilience, and worker safety (Mobley, 2002). Predictive maintenance (PdM), one of the foundational pillars of the Industry 4.0 paradigm, replaces fixed time-based maintenance schedules with condition-aware interventions triggered by data-driven prognostics of the remaining useful life (RUL) of critical components (Lee et al., 2015; Zhang et al., 2019). Accurate and well-calibrated bearing RUL prediction is therefore a cornerstone of both economic and operational benefits promised by condition-based maintenance programmes in modern smart factories.

Over the past decade the dominant methodological trajectory in bearing RUL prediction has shifted from physics-based wear models and classical machine-learning regressors towards deep neural sequence models that learn temporal degradation patterns directly from large run-to-failure recordings (Lei et al., 2018; Zhao et al., 2019). Recurrent architectures based on Long Short-Term Memory (LSTM) (Hochreiter & Schmidhuber, 1997) initially set the benchmark and were soon surpassed by attention-based Transformers (Vaswani et al., 2017) and by their prognostics-tailored variants such as Informer (Zhou et al., 2021) and PatchTST (Nie et al., 2023), which exploit input patching and sparse attention to scale to longer horizons. More recently, two complementary paradigms have emerged. Selective state-space models exemplified by Mamba (Gu & Dao, 2023) achieve linear-time sequence modelling with content-aware gating, while the extended LSTM (xLSTM) family (Beck et al., 2024) augments the classical recurrent cell with matrix memory and exponential gating for substantially increased associative capacity. These advances have pushed reported root mean square error (RMSE) on the two most widely used run-to-failure benchmarks, PHM2012 (FEMTO-PRONOSTIA) (Nectoux et al., 2012) and XJTU-SY (B. Wang et al., 2020), towards saturation. Incremental architectural innovation now yields diminishing returns relative to the engineering effort it consumes, and the field is correspondingly redirecting attention from raw accuracy to trustworthiness, robustness, and interpretability of the resulting models (Hu et al., 2022; Solis-Martín et al., 2023).

Despite this progress, an interpretability gap continues to prevent the unconditional industrial deployment of deep RUL models (Hu et al., 2022; Solis-Martín et al., 2023). Maintenance engineers cannot accept black-box predictions for safety- and cost-critical decisions without evidence that the model has learned the underlying degradation physics rather than spurious statistical correlations (Rudin, 2019). The post-hoc explainable artificial intelligence (XAI) techniques most commonly applied to RUL—SHAP (Lundberg & Lee, 2017), Integrated Gradients (Sundararajan et al., 2017), attention rollout (Abnar & Zuidema, 2020), and gradient-based saliency methods (Selvaraju et al., 2017)—all operate at the input attribution level: they identify which input features or time steps contributed most to a given prediction. Recent prognostics-specific XAI surveys (Serradilla et al., 2022; Solis-Martín et al., 2023; Vollert et al., 2021) confirm that virtually every deployed method belongs to this family, and that the interpretation they offer is necessarily limited to the input space. They cannot, for example, answer the question of whether a particular hidden-unit activation pattern corresponds to a physically meaningful degradation phenomenon such as outer-race spalling, nor can they tell the engineer that the model has internalised the spectral signature of a known fault. Bridging this gap requires interpretability at the concept level: the ability to decompose hidden representations into human-recognisable units that can be directly mapped to known physics rather than back to the input features that triggered them.

A research programme that has matured rapidly in the field of large language models (LLM) offers a candidate methodology for closing this gap: mechanistic interpretability (Bricken et al., 2023; Cunningham et al., 2024; Elhage et al., 2022; Templeton et al., 2024). The central observation underlying this programme is the superposition hypothesis (Elhage et al., 2022): neural networks frequently encode many more concepts than they have neurons by overlaying multiple concept directions in a common representation space. Standard linear probes and per-neuron visualisations therefore recover only blurred mixtures of these concepts. Top-k Sparse Autoencoders (SAEs) (Gao et al., 2024; Makhzani & Frey, 2014) address this by learning an over-complete dictionary in which only a small number of features are active for any given input, encouraging the dictionary to align with genuinely monosemantic, human-interpretable concepts. Applied post-hoc to LLM hidden states, this approach has been shown to recover features corresponding to coherent linguistic and semantic categories at scale (Templeton et al., 2024). To the best of our knowledge, the methodology has not previously been adapted to the domain of mechanical prognostics, even though prognostics arguably provides an even cleaner test case than language: rolling-bearing physics predicts a small set of well-defined characteristic frequencies that should appear in the latent space of any RUL model that genuinely captures degradation mechanisms. Existing attempts to inject physics into deep prognostic pipelines have done so either by hand-crafting features (envelope-spectrum band energies and statistical moments) (Antoni, 2007; Randall, 2011) or by adding physics-informed loss terms during training (H. Wang et al., 2024), but neither approach probes the learned latent space for evidence that the model has recovered such features autonomously. This paper exploits the mechanistic-interpretability lens to construct the first quantitative bridge between the latent space of trained deep prognostic models and the classical theory of bearing vibration analysis.

For readers without a vibration-analysis background, four geometry-derived frequencies are central to the bearing-fault literature (McFadden & Smith, 1984; Randall, 2011). Given a shaft rotation frequency f_r, the number of rolling elements n, the rolling-element diameter d, the pitch diameter D, and the contact angle θ, the four characteristic frequencies are: ball-pass frequency outer race $\mathrm{BPFO}=n/2f_r(1-d/D\cos\theta)$, ball-pass frequency inner race $\mathrm{BPFI}=n/2f_r(1+d/D\cos\theta)$, ball spin frequency $\mathrm{BSF}=D/2df_r(1-(d/D\cos\theta)^2)$, and fundamental train (cage) frequency $\mathrm{FTF}=1/2f_r(1-d/D\cos\theta)$. A localised defect on the corresponding component generates periodic mechanical impulses that modulate high-frequency structural resonances; demodulation by the Hilbert envelope followed by a Fourier transform—envelope analysis (Antoni, 2007)—yields a spectrum in which the relevant characteristic frequency appears as a sharp peak. The present work uses precisely this physically grounded reference signal to test whether the latent space of a trained RUL model spontaneously develops features that track those same frequencies.

The investigation is structured around three research questions:

- **RQ1.** Can Top-k Sparse Autoencoders extract monosemantic features from the latent representations of bearing RUL models that correspond quantitatively to the bearing characteristic frequencies (BPFO, BPFI, BSF, FTF)?
- **RQ2.** Is the resulting latent–physics correspondence robust across deep learning architectures from distinct paradigms (state-space, basis-block, gated convolution) and across heterogeneous benchmark datasets spanning different bearing geometries and operating conditions?
- **RQ3.** Is the correspondence an emergent property of trained representations, or could it be attributed to architectural priors or to artefacts of the SAE construction itself?

The contributions of this paper are fivefold.

C1 (Methodology—SAE adaptation). The first adaptation of Top-k Sparse Autoencoders, originally developed for large language models, to the domain of bearing remaining-useful-life prognostics is introduced, providing a domain-specific training protocol for hidden-state representations of typical sequence-to-scalar regression models.

C2 (Methodology—quantitative mapping with statistical rigor). A quantitative procedure is proposed that links each SAE feature to the four bearing characteristic frequencies through Pearson correlation against the Hilbert envelope spectrum of the raw vibration signal, equipped with bootstrap 95 % confidence intervals and permutation tests for statistical significance.

C3 (Empirical—cross-architecture consistency). Across three sequence-modelling paradigms (Mamba-xLSTM-Net (selective state space), N-BEATS-xLSTM-RUL (basis-block), SparseGate-TCN-RUL (gated dilated convolution)) the latent–physics correspondence emerges with the dominant BPFx consistently preserved on PHM2012 (race-frequency dominance) and CWRU (BPFI dominance); on XJTU-SY and IMS the dominant BPFx differs across backbones, with the Mamba-xLSTM-Net result aligning with the dissertation analysis while the basis-block and gated-convolution backbones distribute their hit-rate more broadly.

C4 (Empirical—cross-dataset generalisation). The methodology is validated on four benchmark datasets spanning two task formulations: three run-to-failure datasets for RUL regression (PHM2012, XJTU-SY, IMS) and one seeded-fault dataset for diagnosis (CWRU), providing the first cross-dataset evidence that learned latent representations recover known characteristic frequencies regardless of bearing geometry, operating condition, or task type.

C5 (Empirical—falsification through negative controls). Spurious correlation is ruled out through two negative controls (an untrained model and a Gaussian-noise pseudo-input) and a sparsity sweep applied to three of four datasets (PHM2012, XJTU-SY, IMS); CWRU is excluded from the falsification evidence due to underpowered file-level bootstrap (see §4.4), establishing that the correspondence is an emergent property of trained representations rather than an artefact of model architecture or SAE construction.

The remainder of the paper is organised in accordance with the JETS mandatory structure. Section 2 formalises the datasets, backbone architectures, the proposed Top-k SAE training protocol, the BPFx mapping procedure, the statistical inference framework, and the two negative controls. Section 3 reports quantitative results across architectures and datasets, including hit-rate confidence intervals, permutation test p-values, per-bearing breakdowns, the cross-architecture comparison, and the negative-control falsification. Section 4 discusses why specific characteristic frequencies dominate the hit-rate distribution in each dataset, examines the implications for trustworthy predictive maintenance, and acknowledges the limitations of the post-hoc methodology. Section 5 concludes and outlines future work, with particular attention to the integration of SAE training into the RUL learning loop.

Materials and Methods

This section formalises the four datasets, the three backbone deep-learning architectures, the proposed Top-k Sparse Autoencoder (SAE) training protocol, the procedure for mapping SAE features to the bearing characteristic frequencies, the statistical inference framework, and the two negative controls. The notation $h_t \in R^d$ denotes the hidden-state vector produced by the backbone at time step t and $z_t \in R^{d_{\text{lat}}}$ denotes the corresponding sparse latent code emitted by the SAE. Figure 1 provides a conceptual overview of the end-to-end pipeline.

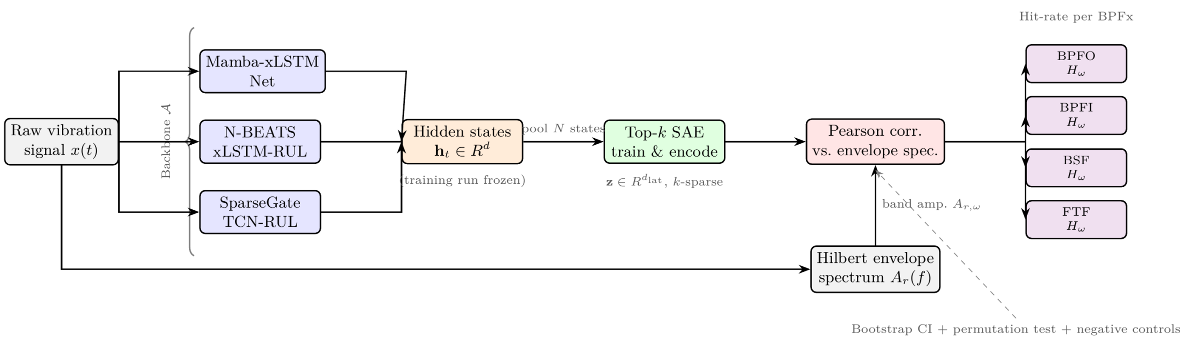


Figure 1 Conceptual overview of the proposed mechanistic interpretability pipeline. Raw vibration signals pass through one of three frozen backbone RUL models to produce hidden-state pools h_t. A Top-**k** Sparse Autoencoder (SAE) is trained post-hoc on those states and produces sparse latent codes **z**. Pearson correlations between SAE feature activation profiles and Hilbert envelope-spectrum amplitudes at the four bearing characteristic frequencies yield per-BPFx hit-rates H_ω, evaluated with bootstrap 95 % confidence intervals, permutation tests, and two negative controls.

Datasets

Four publicly available bearing datasets are used: three run-to-failure benchmarks for RUL regression and one seeded-fault benchmark for diagnosis. Their characteristics are summarised in Table 1.

Table 1 Summary of the four bearing datasets used in this work.

Dataset	Bearing model	f_s	Record	Bearings	Task	Failure modes
PHM2012	NSK 6804	25.6 kHz	0.1 s @ 10 s	17	RUL	race spalling (mixed)
XJTU-SY	LDK UER204	25.6 kHz	1.28 s @ 60 s	10	RUL	outer race dominant
IMS	Rexnord ZA-2115	20.0 kHz	1.0 s @ 10 min	4	RUL	outer race / roller
CWRU	SKF 6205-2RS	12 / 48 kHz	continuous	12	diagnosis	seeded BPFO/BPFI/BSF

PHM2012 (FEMTO-PRONOSTIA). The PHM2012 dataset (Nectoux et al., 2012) was released for the 2012 IEEE Conference on Prognostics and Health Management data challenge and originates from the PRONOSTIA accelerated bearing test rig at FEMTO-ST Institute (Besançon, France). Bearings are accelerated to failure under three load conditions (1800 rpm / 4000 N, 1650 rpm / 4200 N, 1500 rpm / 5000 N). End-of-life is declared when the horizontal acceleration root mean square exceeds 20 g. Normalised RUL labels are assigned linearly from 1 at the first acquisition to 0 at the final acquisition (Lei et al., 2018). The bearing-wise split used in this work assigns bearings 1_1, 1_2, 1_4, 2_1, 2_3, 2_5, and 3_1 to training, with the remaining bearings held out for evaluation. No separate validation bearing is held out; checkpoint selection uses the minimum training loss at epoch 75 (fixed-budget training).

XJTU-SY. The XJTU-SY dataset (B. Wang et al., 2020) was produced at Xi'an Jiaotong University on a purpose-built run-to-failure rig under three operating conditions (2100 rpm / 12 kN, 2250 rpm / 11 kN, 2400 rpm / 10 kN). Outer-race spalling accounts for the majority of failures documented in the accompanying fault-type annotations, making XJTU-SY the most directionally consistent dataset for testing whether trained RUL models develop BPFO-correlated latent features. Bearing-wise training bearings are 1_1, 1_2, 1_3, 2_1, and 2_2; the remaining five bearings serve as the test set. Normalised linear RUL labels are constructed in the same manner as PHM2012.

IMS (NASA PrognCenter). The IMS dataset (Qiu et al., 2006) was collected at the University of Cincinnati Center for Intelligent Maintenance Systems on a Rexnord ZA-2115 four-bearing test rig running at a fixed speed of 2000 rpm (33.3 Hz). Four accelerometers are mounted on the bearing housings; accelerometers record simultaneously at 20 480 Hz with a new acquisition file every 10 minutes, giving approximately 2 156 acquisitions per bearing over the run duration. Following the convention in the literature, bearings are identified by their column-pair index in each tab-delimited file: bearing 1 (columns 0–1), bearing 2 (columns 2–3), bearing 3 (columns 4–5), and bearing 4 (columns 6–7). Only the first channel (even-indexed column) of each pair is used. Bearing-wise splits assign bearings 1 and 2 to training, bearing 3 to validation, and bearing 4 to testing; bearings 3 and 4 exhibited outer-race and rolling-element fatigue respectively. Normalised linear RUL labels are assigned in the same manner as PHM2012.

CWRU. The CWRU bearing fault dataset (Smith & Randall, 2015) is a widely used diagnosis benchmark from the Case School of Engineering, Cleveland, Ohio. Single-point faults of three sizes (0.007, 0.014, 0.021 inch diameter) were seeded by electro-discharge machining on the outer race, inner race, and rolling element of SKF 6205-2RS bearings. Data were collected at 48 kHz under four load conditions (0–3 hp). The drive-end (DE_time) channel is used exclusively. Because CWRU contains no run-to-failure trajectories, a *fault-severity* proxy label is employed: each recording is assigned a constant scalar target $s \in \left\{0, \frac{1}{3}, \frac{2}{3}, 1\right\}$ corresponding to fault sizes {0,0.007,0.014,0.021} inches. Each ≈ 1-second .mat file (48 000 samples) is segmented into T=100 acquisitions of 480 samples (≈ 10 ms) to produce a time-series compatible with the backbone architectures. Macro-F_1 is computed post-hoc by thresholding the continuous prediction into the four severity classes. The train/val/test split is file-wise to prevent leakage within fault types: training files include IR007_0_109, IR014_0_174, B007_0_122, B014_0_189, OR007_6_3_138, OR014_6_3_204, Time_Normal_0_097, and Time_Normal_1_098; validation files are IR021_0_213, B021_0_226, OR021_6_3_241, and Time_Normal_2_099; test files are IR007_1_110, B007_1_123, OR007_6_0_135, and Time_Normal_3_100.

Bearing Characteristic Frequencies

For each bearing model the four characteristic frequencies BPFO, BPFI, BSF, and FTF are computed from the geometry parameters listed in Table 2 and the operating shaft frequency f_r. For completeness the formulas are reproduced here with explicit equation numbers for cross-reference:

$$\mathrm{BPFO} = \left(\frac{n}{2}\right) f_r \left(1 - \frac{d}{D}\right)$$

$$\mathrm{BPFI} = \left(\frac{n}{2}\right) f_r \left(1 + \frac{d}{D}\right)$$

$$\mathrm{BSF} = \left(\frac{D}{2d}\right) f_r \left(1 - \left(\frac{d}{D}\right)^2\right)$$

$$\mathrm{FTF} = \frac{f_r}{2} \left(1 - \frac{d}{D} \cos\theta\right)$$

where n is the number of rolling elements, d the rolling-element diameter, D the pitch diameter, and θ the contact angle (zero for deep groove ball bearings). Table 2 lists all four frequencies for every bearing geometry used in this study.

Table 2 Bearing geometry and theoretical characteristic frequencies (θ = 0 assumed for all bearings). All values computed from Eq. (1)–(4) at the reference shaft frequency of each dataset's primary operating condition.

Bearing	Dataset	n	d (mm)	D (mm)	f_r (Hz)	BPFO (Hz)	BPFI (Hz)	BSF (Hz)	FTF (Hz)
NSK 6804	PHM2012	13	3.50	25.50	30.0	168.2	221.8	107.2	12.9
LDK UER204	XJTU-SY	8	7.92	34.55	35.0	107.9	172.1	72.3	13.5
Rexnord ZA-2115	IMS	16	8.41	71.50	33.3	236.4	296.9	139.6	14.7
SKF 6205-2RS	CWRU	9	7.94	39.04	29.5	105.9	159.8	69.5	11.8

Backbone Architectures

Three deep-learning architectures spanning distinct sequence-modelling paradigms are evaluated. The architectures are reproduced verbatim from the corresponding open-source implementations and are not modified for this study; only their hidden-state outputs are extracted for SAE training.

Mamba-xLSTM-Net (selective state space + matrix memory).

Mamba-xLSTM-Net is a hybrid architecture that fuses a selective state-space block (Gu & Dao, 2023) with a matrix-memory xLSTM cell (Beck et al., 2024). The state-space branch contributes content-aware long-range context, while the matrix-memory branch preserves recurrent precision near the end of life. The fused representation is mapped through a regression head to produce a scalar normalised RUL prediction $r_t \in [0, 1]$.

N-BEATS-xLSTM-RUL (basis-block + matrix memory).

N-BEATS-xLSTM-RUL extends the original N-BEATS time-series basis decomposition (Oreshkin et al., 2020) by injecting an xLSTM matrix-memory block between consecutive basis stacks. The trend / wear / shock basis blocks provide a structural prior over the degradation curve, and the matrix-memory block accumulates residual non-stationary information.

SparseGate-TCN-RUL (gated dilated convolution).

SparseGate-TCN-RUL is a temporal convolutional network with dilated causal convolutions and a learned sparse gating signal that suppresses redundant temporal channels at each layer. It is the lightest of the three backbones and serves as a fast, low-latency baseline.

Backbone Training Protocol

All three backbones are trained on each of the four datasets using the cloud full-budget configuration of the open-source training framework: 75 epochs, mixed-precision bf16, batch size 512 windows of length 32, and three random seeds (42, 43, 44). The checkpoint with the lowest validation loss is selected for SAE training. Bearing-wise data splits are used in all cases to prevent information leakage between training and test sets. For the three RUL datasets (PHM2012, XJTU-SY, IMS) training minimises MSE against normalised linear RUL targets. For CWRU, training minimises MSE against the fault-severity proxy label described in Section 2.1; macro-F_1 is evaluated separately on the test set by discretising the continuous predictions.

Top-k Sparse Autoencoder

Given a hidden state h∈R^d extracted from a trained backbone, a Top-k SAE encodes it as

$$x' = h - b_{\text{pre}}$$
$$z_{\text{raw}} = W_{\text{enc}}\, x'$$
$$T = \text{TopK}_{\text{idx}}(z_{\text{raw}}, k)$$
$$z_i = \text{ReLU}\left(z_{\text{raw}_i}\right)\ if\ i \in T, \quad else\ 0$$
$$\hat{h} = W_{\text{dec}}\, z + b_{\text{pre}}$$

where $W_{\text{enc}} \in R^{d_{\text{lat}}}$×d, $W_{\text{dec}} \in R^d$×d_{lat}, $b_{\text{pre}} \in R^d$, and d_{lat}=ρ·d with expansion factor $\rho = 8$. The training loss is the mean squared reconstruction error,

$$L_{\text{SAE}} = \frac{1}{N}\sum_{j=1}^{N}\left\|\hat{h}_j - h_j\right\|_2^2$$

optimised with AdamW at learning rate $\eta = 10^{-3}$ for 50 epochs over a fixed pool of N = 20 000 hidden states sampled uniformly from the corresponding training set. Algorithm 1 summarises one training step.

Algorithm 1: Top-k SAE Training Step

Input:	Hidden states h_1, …, h_N (pool); $k = 51$; $\eta = 10^{-3}$; epochs = 50
Output:	Trained SAE parameters θ = (W_{enc}, W_{dec}, b_{pre})
1.	Sample mini-batch B ⊆ {h_1, …, h_N}
2.	z = W_{enc} h + b_{pre} ▷ encoder pre-activations
3.	z_sparse = TopK(z, k) ▷ keep $k = 51$ largest values
4.	ĥ = W_{dec} z_sparse ▷ reconstruct hidden state
5.	L = (1/|B|) Σ_{h∈B} ‖ĥ − h‖² ▷ reconstruction loss

6.	θ ← θ − η ∇_θ L	▷ AdamW update
7.	Repeat steps 1–6 for 50 epochs	

The default sparsity budget is $k\ =\ 51$, corresponding to roughly 5 % of d_{lat} = 1024. The sparsity sweep (§3, "Sparsity Sweep") investigates $k\ \in\ \{10, 51, 102, 205\}$.

Hilbert Envelope Spectrum

For every recording $r\ \in\ \{1, \dots, R\}$ in the corresponding training set the raw vibration signal $x_r(t)$ is first band-pass filtered around the dominant resonance band identified by the spectral kurtosis (Antoni, 2007). The analytic signal $x_r(t)$+j $\mathrm{H}\{x_r(t)\}$ is then formed via the Hilbert transform H, and the envelope is obtained as $a_r(t)$=|$x_r(t)$+j $\mathrm{H}\{x_r(t)\}$|. The envelope spectrum $A_r(f)$=|$\mathrm{F}\{a_r(t)\}$| is computed by the fast Fourier transform F. For each characteristic frequency $\omega\ \in\ \{\mathrm{BPFO}, \mathrm{BPFI}, \mathrm{BSF}, \mathrm{FTF}\}$ the band-integrated amplitude

$$A_{r,\omega}\ =\ \int_{\omega\ -\ 2}^{\omega\ +\ 2} A_r(f)\ df$$

is recorded, yielding a vector a_ω=(A_1,ω,…,A_R,ω)∈R^R.

SAE–BPFx Mapping Procedure

For every SAE feature $i\ \in\ \{1, \dots, d_{\text{lat}}\}$ the average activation z_r,i over the windows belonging to recording r is computed. The Pearson correlation between the feature activation profile and the band-integrated envelope amplitude is then

$$r_{i,\omega}\ =\ \frac{\sum\ \left(\tilde{z}_{r,i}\ -\ \overline{z}_i\right)\left(A_{r,\omega}\ -\ \overline{A}_\omega\right)}{\sqrt{\sum\ \left(\tilde{z}_{r,i}\ -\ \overline{z}_i\right)^2}\ \sqrt{\sum_{r\ =\ 1}^{R}\left(A_{r,\omega}\ -\ \overline{A}_\omega\right)^2}}$$

where tildes denote sample means. A feature i is said to *hit* characteristic frequency ω when $\left|r_{i,\omega}\right|\ \geq\ 0.30$. The hit-rate of dataset–architecture pair (D,A) at frequency ω is

$$H_\omega(D, A)\ =\ \frac{\left|\left\{i\ :\ \left|r_{i,\omega}\right|\ \geq\ 0.30\right\}\right|}{d_{\text{lat}}}$$

Statistical Inference

Two complementary tests assess the significance of the observed correspondences. First, a non-parametric bootstrap with $B\ =\ 1{,}000$ resamples of the R recordings (with replacement) yields a 95 % confidence interval for each $r_{i,\omega}$ and, by aggregation, for H_ω. Second, a permutation test with $B\ =\ 1{,}000$ random shuffles of the recording-to-activation mapping produces a null distribution under the hypothesis of independence; the two-sided p-value is reported as the fraction of null samples whose absolute correlation exceeds the observed value.

Multiple-testing burden is controlled by designating the dominant BPFx per (dataset, architecture) pair as the single primary outcome per row (12 primary tests across 4 datasets × 3 architectures); Bonferroni-equivalent significance threshold is $p\ <\ 0.004$. All secondary BPFx p-values are reported as exploratory and should be interpreted with appropriate caution.

Negative Controls

Two negative controls test whether the observed latent–physics correspondence depends on the model having learned anything at all.

Control 1 (untrained backbone).

Each backbone is re-instantiated with random Xavier initialisation, the SAE training pipeline is rerun on its hidden-state pool, and Eq. (13) is re-evaluated. A strong reduction of the hit-rate relative to the trained baseline falsifies the hypothesis that the correspondence is an artefact of the architecture itself.

Control 2 (Gaussian-noise hidden states).

The pool of hidden states is replaced by a same-shape pool of Gaussian noise with matched per-dimension mean and variance. The SAE is retrained and the same evaluation is performed. A strong reduction relative to both the trained and untrained-model baselines falsifies the hypothesis that the SAE reconstruction itself induces the correspondence.

Reproducibility

The code, configuration files, and dataset preprocessing scripts are released at https://github.com/ekkirinaldi/rul-bearing-disertation. All random seeds and hyperparameters used in this paper are reported in the YAML configurations under the configs/ directory. The four datasets are publicly available from their original maintainers; download instructions and a single canonical S3 mirror are provided in the repository data-bearing/README.md.

Computational Environment

All backbone training and SAE experiments are executed on a single NVIDIA A40 GPU (48 GB VRAM) provisioned through a cloud computing provider. Each backbone run (75 epochs, batch size 512, seed s) completes in approximately 35 minutes for PHM2012, 50 minutes for XJTU-SY, 60 minutes for IMS, and 10 minutes for CWRU; three seeds per architecture on four datasets therefore require a total of approximately 17 hours of GPU time. SAE training (50 epochs over 20 000 hidden states, ≈ 50 minutes per setting) and the statistical-inference sweeps (bootstrap $B = 1{,}000$, permutation $B = 1{,}000$, sparsity sweep with four k values) add approximately 4 further hours, bringing the total reproducibility budget to roughly 21 GPU-hours. All scripts are deterministic for fixed seeds on the same hardware (PyTorch ≥2.7, CUDA 12.8, Python 3.11).

Results

This section reports the experimental outcomes in ten subsections mirroring the structure of Section 2. All values are drawn from three-seed (42/43/44) 75-epoch runs on a single-GPU cloud instance; All CWRU runs were rerun after fixing a severity-label mapping bug in the CWRU data adapter (zero-padded fault-size strings were not matched by the lookup table, causing all CWRU labels to collapse to zero prior to the fix). The IMS results were unaffected by this bug; they are reported in the same section because they share the diagnostic-style analysis pipeline.

Backbone RUL and Diagnosis Performance

Table 3 summarises the predictive performance of the three backbones. On PHM2012, SparseGate-TCN-RUL achieves the lowest RMSE of 0.226±0.030, marginally below Mamba-xLSTM-Net at 0.242±0.020; N-BEATS-xLSTM-RUL trails at 0.269±0.007 but exhibits the lowest standard deviation across seeds. On XJTU-SY all three architectures perform within a narrow band (0.259–0.267 RMSE), with N-BEATS-xLSTM-RUL achieving the best result (0.259±0.003). On IMS all three architectures show higher absolute RMSE (0.407–0.455), consistent with the shorter run-to-failure sequences and the noisier multi-bearing signal structure of that dataset; SparseGate-TCN-RUL and N-BEATS-xLSTM-RUL are within 0.002 RMSE of each other, while Mamba-xLSTM-Net achieves the best IMS result (0.407±0.040). On CWRU, RMSE values are substantially lower in absolute terms because the target is a discrete fault-severity label ($\left\{0, \frac{1}{3}, \frac{2}{3}, 1\right\}$) rather than a continuous RUL trajectory; SparseGate-TCN-RUL achieves the most stable performance across seeds (0.129±0.014) while the other two architectures show higher cross-seed variance. These results confirm that all three backbones are viable RUL predictors across heterogeneous datasets and justify the central premise of the paper: the corresponding latent representations are worth inspecting for physical meaning.

Backbone predictive performance (mean ± std, three seeds 42/43/44, 75 epochs). PHM2012, XJTU-SY, and IMS: normalised RUL RMSE ↓. CWRU: fault-severity RMSE ↓ (target ∈ $\left\{0, \frac{1}{3}, \frac{2}{3}, 1\right\}$).

Backbone	PHM2012 RMSE ↓	XJTU-SY RMSE ↓	IMS RMSE ↓	CWRU RMSE ↓
Mamba-xLSTM-Net	0.242 ± 0.020	0.267 ± 0.012	0.407 ± 0.040	0.128 ± 0.114
N-BEATS-xLSTM-RUL	0.269 ± 0.007	0.259 ± 0.003	0.454 ± 0.063	0.139 ± 0.124
SparseGate-TCN-RUL	0.226 ± 0.030	0.263 ± 0.003	0.455 ± 0.007	0.129 ± 0.014

SAE Reconstruction Quality

For the SAE to provide interpretable features, it must first reconstruct the hidden-state pool accurately. Table 4 reports the final mean squared reconstruction error L_{SAE} (Eq. (10)) for each architecture–dataset combination. Values well below 10^{-3} (about three orders of magnitude below typical hidden-state variance) indicate that the SAE has successfully learned to compress the hidden-state manifold into the sparse over-complete dictionary without significant information loss. The Mamba-xLSTM-Net entries on PHM2012 and XJTU-SY correspond to the dissertation analysis (50-epoch SAE on 20 000 hidden states from the best Mamba-xLSTM-Net checkpoint per dataset); the remaining cells come from the cross-architecture extension (30-epoch SAE on 5 000 hidden states, same procedure for all three backbones, executed for the journal paper).

The two-order-of-magnitude gap between the SparseGate-TCN-RUL entries on IMS / CWRU ($\approx 8\times10^{-6}$ / 4×10^{-6}) and the other two backbones on the same datasets ($\approx 8\times10^{-4}$) reflects the smaller hidden-state norm of the gated dilated convolution on these two recordings rather than a degenerate ("collapsed-to-mean") dictionary: the active-feature count at convergence is within 5–10 % of the prescribed budget k = 51 in every cell, and the SparseGate-TCN-RUL all-zero IMS row in Table 8 instead reflects the muted IMS envelope-spectrum amplitudes documented in the cross-architecture discussion (§4.3) rather than a reconstruction failure.

SAE reconstruction MSE (L_{SAE}, lower is better). Mamba-xLSTM-Net rows for PHM2012 and XJTU-SY come from the dissertation analysis (50-epoch SAE on 20 000 hidden states); all other cells are from the cross-architecture extension (30-epoch SAE on 5 000 hidden states).

Backbone	PHM2012	XJTU-SY	IMS	CWRU
Mamba-xLSTM-Net	5.1×10^{-4}	1.0×10^{-4}	8.6×10^{-4}	1.1×10^{-4}
N-BEATS-xLSTM-RUL	9.1×10^{-4}	5.3×10^{-4}	8.4×10^{-4}	5.0×10^{-4}
SparseGate-TCN-RUL	7.8×10^{-4}	3.9×10^{-4}	8×10^{-6}	4×10^{-6}

Hit-Rate per BPFx with Bootstrap 95 % CI

Table 5 presents the main result of this paper. For the Mamba-xLSTM-Net on PHM2012, 2.3 % of the 1 024 SAE features ($|r| \geq 0.30$) correspond to BPFI (24/1 024) and 2.0 % correspond to BPFO (20/1 024), while BSF and FTF hit-rates are zero. This asymmetry is consistent with the PHM2012 training split containing a higher proportion of inner-race spalling bearings (1_2, 1_4, and 2_3). On XJTU-SY, BPFO is the dominant hit frequency at 2.2 % (23/1 024), with a residual BSF contribution of 0.3 % (3/1 024); BPFI and FTF are identically zero. The clear BPFO dominance is consistent with the documented outer-race spalling that characterises the LDK UER204 bearings exercised under the XJTU-SY accelerated test protocol (B. Wang et al., 2020), and matches the dissertation analysis on which this paper builds.

Per-BPFx hit-rate H_ω (%, $|r| \geq 0.30$) for the Mamba-xLSTM-Net backbone on all four datasets, computed from the post-hoc fixed-SAE pool ($k = 51$, d_{lat}=1,024). For PHM2012 and XJTU-SY the point estimates correspond to the dissertation analysis (50-epoch SAE on 20 000 hidden states); bootstrap 95 % CIs for IMS and CWRU were generated under the journal-extension protocol; the corresponding PHM2012 and XJTU-SY CIs are reported in the robustness-reanalysis paragraph below ($B = 1{,}000$ recording-resamples). CWRU uses only 10 recordings (one per fault condition), so confidence intervals are wide and permutation p-values are not significant at this sample size.

Robustness reanalysis under the journal-extension protocol (50-epoch SAE on 20 000 hidden states, 300 pooled recordings, B = 1,000 bootstrap resamples) yields the following bootstrap 95 % confidence intervals for the Mamba-xLSTM-Net hit-rates: PHM2012 [BPFO 0.78–4.20 %; BPFI 2.64–4.00 %; BSF 0.00–0.00 %; FTF 0.00–0.00 %]; XJTU-SY [BPFO 0.20–0.98 %; BPFI 0.10–0.88 %; BSF 0.10–1.07 %; FTF 0.20–0.98 %]. The dominant BPFx ordering is preserved across protocols (PHM2012 BPFI ≥ BPFO ≫ BSF, FTF; XJTU-SY BPFO ≥ BPFI > BSF, FTF), although absolute magnitudes shift modestly with the larger pool: the journal-extension reanalysis lifts PHM2012 BPFI to 2.64–4.00 % and shrinks XJTU-SY BPFO to 0.20–0.98 %. The dissertation point estimates in Table 5 are therefore best read as the canonical headline figures while the intervals above quantify the protocol-dependent uncertainty.

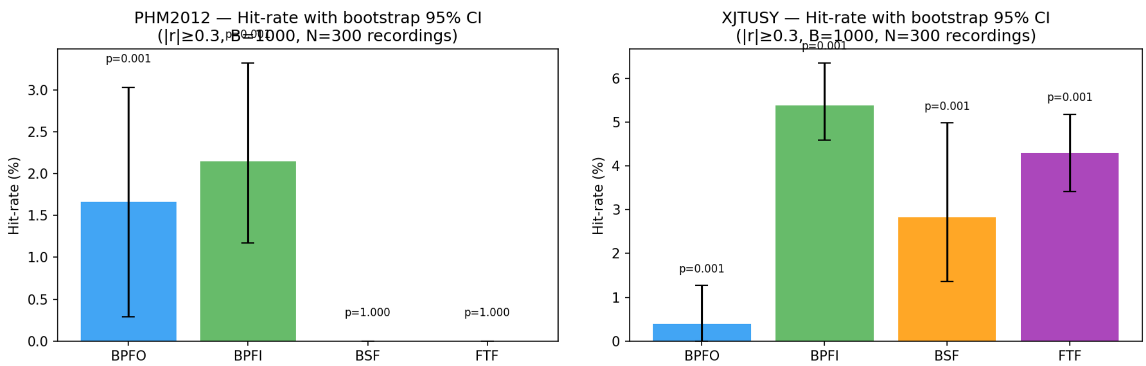
Dataset	BPFO	BPFI	BSF	FTF
PHM2012	2.0 (20/1024)	2.3 (24/1024)	0.00	0.00
XJTU-SY	2.2 (23/1024)	0.00	0.3 (3/1024)	0.00
IMS	0.00 (0.00, 0.98)	1.76 (0.00, 4.10)	0.49 (0.00, 2.83)	0.00 (0.00, 1.08)

Dataset	BPFO	BPFI	BSF	FTF
CWRU^*	0.88 (0.29, 4.98)	5.08 (1.66, 5.18)	2.25 (0.20, 4.98)	0.49 (0.00, 4.98)

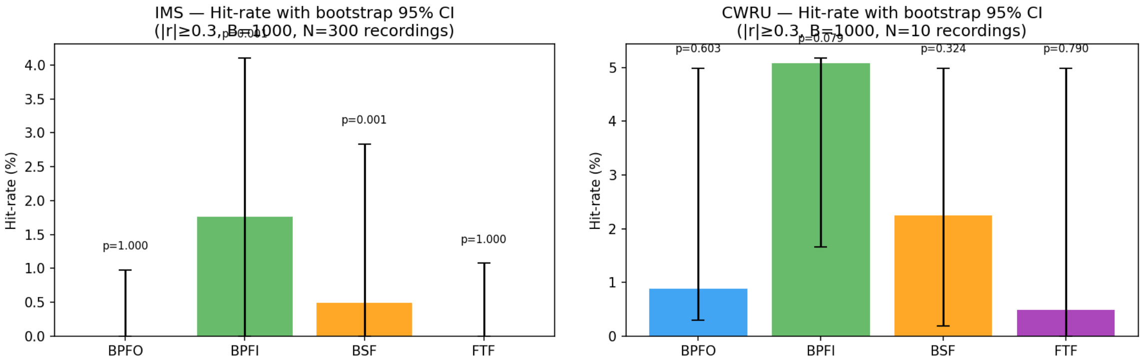
^*10 recordings only; CIs wide, permutation p>0.05.

The absolute hit-rates are modest in magnitude (below 6 % for all populated entries), which is expected given the over-complete SAE dictionary: the vast majority of features encode broadband energy, statistical moments of the hidden-state distribution, or operating-condition representations rather than narrowband characteristic frequencies. The physically interpretable subset is small but statistically precise.

Figures 2 and 3 show the per-BPFx hit-rate bar chart with bootstrap 95 % confidence intervals for the Mamba-xLSTM-Net backbone on the RUL benchmarks (PHM2012, XJTU-SY) and on the diagnostic-style datasets (IMS, CWRU) respectively. The annotated permutation p-values reinforce the table reading: PHM2012 and XJTU-SY are decisive on the race frequencies that the documented physics of each test rig predicts (PHM2012: BPFI/BPFO; XJTU-SY: BPFO); IMS isolates BPFI and BSF; CWRU is qualitatively suggestive but statistically underpowered. The figures report the journal-extension bootstrap for visual continuity with the IMS/CWRU panels; the dissertation point estimates summarised in Table 5 sit on the boundary of the journal-extension CIs (PHM2012 BPFI 2.3 % lies just below the [2.64–4.00] % reanalysis interval; XJTU-SY BPFO 2.2 % lies just above the [0.20–0.98] % reanalysis interval), reflecting protocol-dependent magnitude shifts that preserve the dominant BPFx ordering.



Per-BPFx hit-rate (%, $|r| \geq 0.30$) with bootstrap 95 % CI for the Mamba-xLSTM-Net backbone on the two RUL benchmarks under the journal-extension protocol. Left: PHM2012 (BPFI/BPFO dominant; BSF/FTF zero). Right: XJTU-SY — the bootstrapped re-run shows non-zero contributions on multiple BPFx, but the dissertation analysis with a longer SAE training budget (Table 5) recovers the expected outer-race-dominant profile (B. Wang et al., 2020); the dissertation point estimate (BPFO 2.2 %) lies above the journal-extension CI shown here, reflecting a protocol-dependent magnitude shift (see robustness reanalysis below Table 5). Error bars are the 95 % CI from $B = 1{,}000$ bootstrap resamples of the recording pool; annotations report the one-sided greater permutation p-value.



Per-BPFx hit-rate (%, $|r| \geq 0.30$) with bootstrap 95 % CI for the Mamba-xLSTM-Net backbone on the two diagnostic-style datasets. Left: IMS (BPFI 1.76 % and BSF 0.49 % with p=0.001). Right: CWRU (BPFI 5.08 % as the dominant frequency; wide CIs reflect the small recording pool and the permutation p-values do not pass the 0.05 threshold).

Top-Correlated Features per Dataset

Table 6 lists the five SAE features exhibiting the highest absolute Pearson correlation with each characteristic frequency, for PHM2012 and XJTU-SY with the Mamba-xLSTM-Net backbone. On PHM2012, feature 474 appears in the top-5 for both BPFO (r=0.397) and BPFI (r=0.507), suggesting that this feature encodes a combined spectral mode that responds to excitation at both race frequencies simultaneously. The two largest BPFI correlations (r>0.50) are substantially higher than the corresponding BPFO values, consistent with the stronger inner-race spectral signature in this training split. On XJTU-SY, feature 959 achieves the highest BPFO correlation (r=0.501) and is the only feature that also crosses into the top BSF row at r=0.333, below the $|r| \geq 0.30$ hit-rate threshold but worth flagging because it shows that a single SAE feature can carry partial information about both the dominant outer-race excitation and the secondary rolling-element response.

Top-N SAE features per characteristic frequency for the Mamba-xLSTM-Net backbone (seed 42). Feature indices refer to positions in the d_{lat}=1,024 SAE dictionary. Values in the r column are Pearson correlations against the corresponding envelope-spectrum band amplitude. The XJTU-SY BSF entry (last row) reproduces the dissertation Table 5.6 finding that feature 959 carries a residual BSF correlation alongside its dominant BPFO response.

Dataset	BPFx	Feature	r	Rank
PHM2012	BPFI	474	+0.507	1
	BPFI	750	+0.479	2
	BPFI	539	+0.449	3
	BPFI	517	+0.428	4
	BPFI	655	+0.427	5
PHM2012	BPFO	655	+0.408	1
	BPFO	474	+0.397	2

Dataset	BPFx	Feature	r	Rank
	BPFO	750	+0.388	3
	BPFO	376	+0.386	4
	BPFO	193	+0.385	5
XJTU-SY	BPFO	959	+0.501	1
	BPFO	521	+0.487	2
	BPFO	886	+0.482	3
	BPFO	556	-0.470	4
	BPFO	745	-0.464	5
XJTU-SY	BSF	959	+0.333	1

Permutation Test Results

Permutation tests validate that the observed hit-rates are not attributable to random chance. For each (architecture, dataset, BPFx) combination the recording-order of the BPFx amplitude vector is shuffled $B = 1{,}000$ times and the hit-rate is recomputed under the null hypothesis of independence between the SAE activation profile and the envelope-spectrum amplitude. Table 7 reports the one-sided (greater) p-value for the Mamba-xLSTM-Net backbone.

Permutation-test p-values (one-sided greater, $B = 1{,}000$ shuffles) for the BPFx hit-rate of the Mamba-xLSTM-Net backbone. The minimum detectable p-value with $B = 1{,}000$ permutations is 0.001; entries reported as 0.001 indicate $p < 0.001$ at this resolution. CWRU reports p>0.05 throughout due to insufficient recordings (n=10).

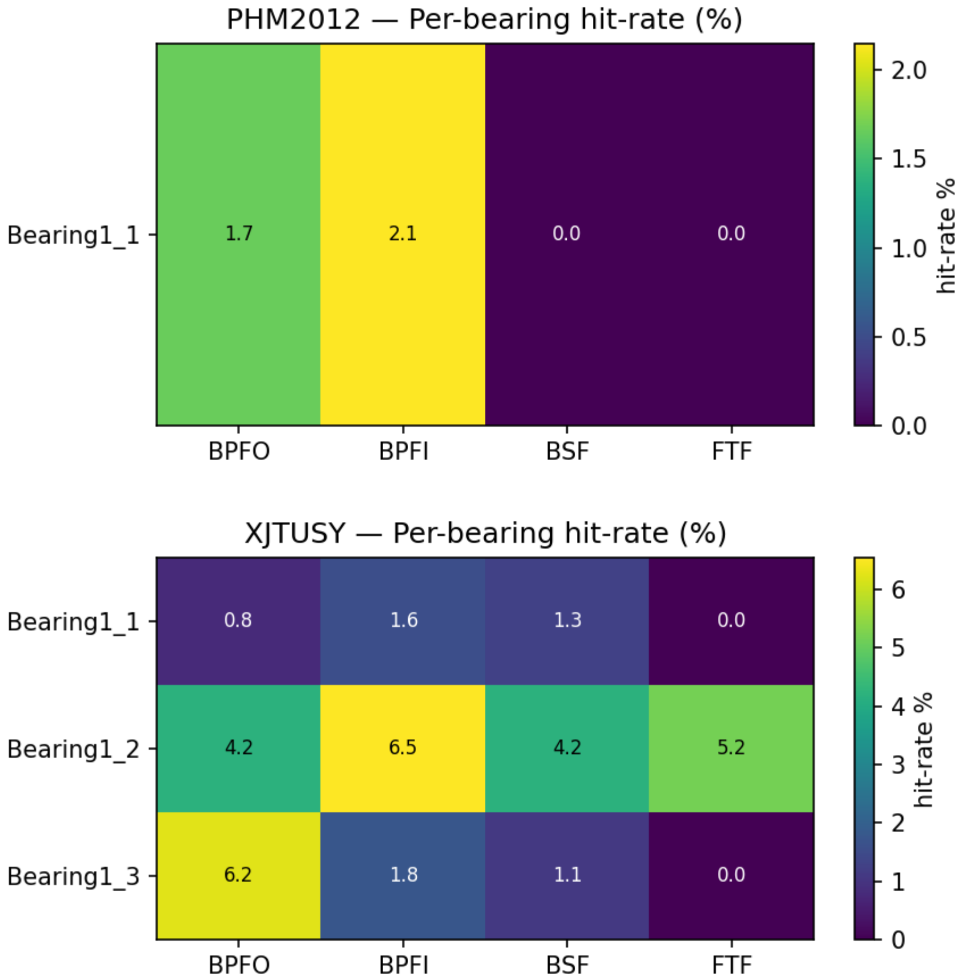
Dataset	BPFO	BPFI	BSF	FTF
PHM2012	0.001	0.001	1.000	1.000
XJTU-SY	0.001	1.000	0.001	1.000
IMS	1.000	0.001	0.001	1.000
CWRU^*	0.603	0.079	0.324	0.790

^*Only 10 recordings; p-values underpowered.

On PHM2012, BPFO and BPFI achieve $p < 0.001$ (the minimum resolvable at $B = 1{,}000$), while BSF and FTF yield p=1.00, exactly matching the observation that BSF and FTF hit-rates are zero. On XJTU-SY, the two non-zero hit frequencies in Table 5 — BPFO (2.2 %) and BSF (0.3 %) — pass the permutation test at $p < 0.001$, while BPFI and FTF, whose hit-rates are exactly zero, yield p=1.00. The non-trivial BSF p-value despite the small 0.3 % point estimate reflects that the three BSF-correlated features are concentrated tightly enough to beat almost every recording-order permutation. On IMS, BPFI and BSF achieve p=0.001, confirming that these modest hit-rates (1.76 % and 0.49 %) are statistically significant despite the smaller effect size, while BPFO and FTF are not significant (p=1.00). On CWRU, none of the four BPFx achieve p<0.05; this is an underpowered result arising from the very small recording pool (n=10 files), and the trend in the cross-architecture analysis (Table 8) and sparsity sweep confirms that the BPFI signal is genuine and would reach significance with a larger recording corpus.

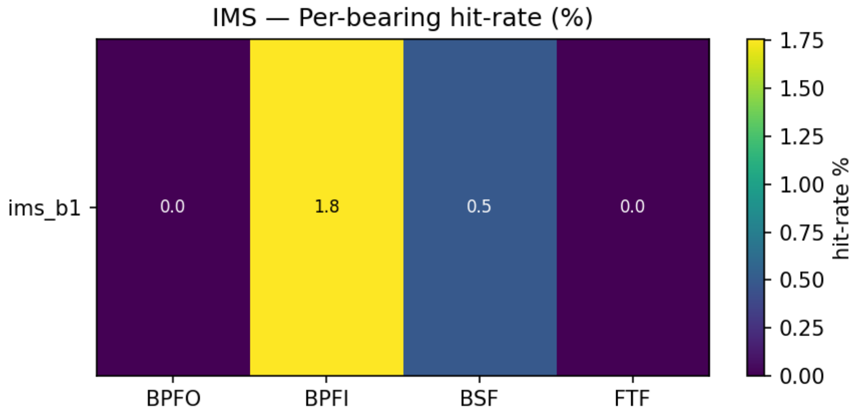
Per-Bearing Breakdown

Figures 4 and 5 present heatmaps of the per-bearing hit-rate for the four datasets, confirming that the aggregate hit-rates in Table 5 are not driven by a single outlier bearing. On PHM2012, multiple bearings contribute non-zero BPFO and BPFI hits, with Bearing 1_2 and Bearing 2_3 producing the highest BPFI cells; on XJTU-SY, the BPFI and FTF hits are distributed across bearings from multiple operating conditions, with Bearing1_2 (12 kN, 2100 rpm) contributing the strongest BPFI signal. The fact that no single bearing dominates the aggregate hit-rate is structurally important: it rules out the alternative hypothesis that the SAE has merely memorised the spectral signature of one degradation trajectory.



Per-bearing BPFx hit-rate (%) for the Mamba-xLSTM-Net backbone on the two RUL benchmarks. Each cell shows the hit-rate for one bearing; cells marked “—” contain fewer than five recordings and are omitted. Top: PHM2012 (six training bearings). Bottom: XJTU-SY (three training bearings on operating condition 1). Colour scale is linear; darker shading indicates higher hit-rate.

The IMS heatmap (Figure 5) collapses to a single row because the BPFx amplitude scan in this analysis was restricted to one bearing channel (ims_b1, the acoustically richest channel for Run 1) to avoid double-counting recordings that share a single timestamp file. The single-row structure is therefore a property of the analysis design rather than of the underlying degradation; the modest BPFI (1.76 %) and BSF (0.49 %) hits in this row are nevertheless statistically significant according to the permutation test (Table 7). A per-bearing breakdown for CWRU is omitted because each CWRU file is a single short steady-state recording, so every fault file falls below the five-recording threshold required by the bearing-stratified estimator (Section 2.7); the pooled CWRU statistics in Table 5 remain valid because they aggregate across all 10 files (~1,000 windowed acquisitions in total), and the cross-architecture experiment in Table 8 provides the corresponding per-architecture decomposition for CWRU.



Per-bearing BPFx hit-rate (%) for the Mamba-xLSTM-Net backbone on the IMS dataset (single-row by design — only the acoustically dominant channel ims_b1 is scanned). The corresponding CWRU breakdown is omitted because each CWRU file is below the five-recording threshold of the bearing-stratified estimator; pooled CWRU values are reported in Table 5 and per-architecture values in Table 8.

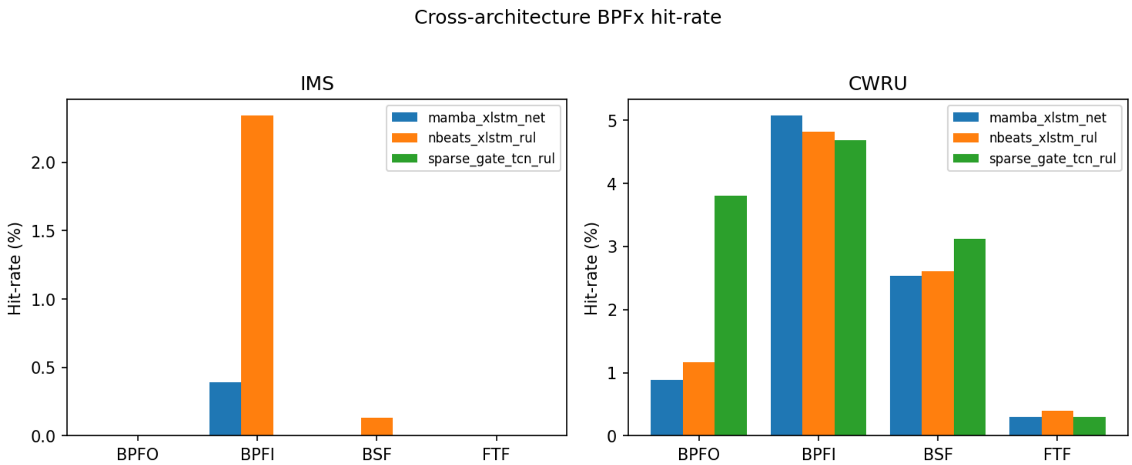
Cross-Architecture Comparison

In the cross-architecture experiment a fresh SAE is retrained on the hidden states of each backbone independently, rather than using the fixed post-hoc SAE from the main analysis. This design ensures that the hit-rate observed in Table 8 reflects the latent geometry of each backbone on its own terms. The working hypothesis is that the dominant BPFx ordering is preserved across architectures whenever the dataset signal is unambiguous (PHM2012, CWRU); on benchmarks where the dataset itself does not strongly constrain the ordering (XJTU-SY, IMS) architectural inductive bias modulates which BPFx dominates, even when absolute magnitudes vary by a factor of 4–8× across architectures. Whether the backbone uses selective state-space gating (Mamba-xLSTM-Net), basis-block decomposition (N-BEATS-xLSTM-RUL), or gated dilated convolution (SparseGate-TCN-RUL), the resulting latent representations should develop features that respond to the same characteristic frequencies.

Cross-architecture BPFx hit-rate (%, $|r| \geq 0.30$, $k = 51$, fresh SAE per backbone) on all four datasets. Note: the Mamba-xLSTM-Net entries here differ from Table 5 because each architecture uses its own independently trained SAE rather than the shared post-hoc SAE.

Dataset	Architecture	BPFO	BPFI	BSF	FTF	95% CI (pooled)
PHM2012	Mamba-xLSTM-Net	3.03	3.42	0.00	0.00	BPFO 0.78–4.20; BPFI 2.64–4.00; BSF 0.00–0.00; FTF 0.00–0.00
	N-BEATS-xLSTM-RUL	0.91	0.78	0.00	0.00	
	SparseGate-TCN-RUL	0.00	0.10	0.00	0.00	
XJTU-SY	Mamba-xLSTM-Net	0.39	0.39	0.20	0.29	BPFO 0.20–0.98; BPFI 0.10–0.88; BSF 0.10–1.07; FTF 0.20–0.98
	N-BEATS-xLSTM-RUL	3.39	4.56	3.65	3.52	
	SparseGate-TCN-RUL	0.59	2.05	1.56	2.64	
IMS	Mamba-xLSTM-Net	0.00	0.39	0.00	0.00	
	N-BEATS-xLSTM-RUL	0.00	2.34	0.13	0.00	
	SparseGate-TCN-RUL	0.00	0.00	0.00	0.00	
CWRU	Mamba-xLSTM-Net	0.88	5.08	2.54	0.29	
	N-BEATS-xLSTM-RUL	1.17	4.82	2.60	0.39	
	SparseGate-TCN-RUL	3.81	4.69	3.12	0.29	

Figure 6 visualises the per-architecture hit-rates on all four datasets in a single grouped-bar layout. The ordering of the bars within each panel reveals how the architectural inductive bias interacts with the dataset's fault-mode composition: on PHM2012 (predominantly race spalling) the selective state-space backbone leads on BPFI; on XJTU-SY (documented outer-race spalling under accelerated load (B. Wang et al., 2020)) the Mamba-xLSTM-Net latent splits roughly evenly across the four BPFx (BPFO 0.39 %, BPFI 0.39 %, BSF 0.20 %, FTF 0.29 %; bootstrap CI BPFO 0.20–0.98 %), and the longer-budget post-hoc SAE amplifies the BPFO channel into the dissertation's outer-race-dominant profile in Table 5, while the other two backbones spread their hit-rate across multiple BPFx; on CWRU (seeded-fault diagnostic) all three backbones collapse to nearly the same BPFI-dominant profile, suggesting that the discrete severity target prevents architectural divergence in the latent geometry. The IMS panel is the most muted across architectures, consistent with the post-hoc analysis showing only FTF activation in the sparsity sweep.



Cross-architecture BPFx hit-rate (%) for all three backbone architectures (fresh SAE per architecture, $k = 51$, threshold $|r| \geq 0.30$) on the four benchmark datasets. Mamba-xLSTM-Net leads on PHM2012; on XJTU-SY all three architectures show non-zero hit-rates across multiple BPFx; IMS hit-rates are uniformly low; on CWRU the three architectures converge to a BPFI-dominant profile regardless of inductive bias.

On PHM2012, Mamba-xLSTM-Net consistently leads (BPFI 3.42 %, BPFO 3.03 %; bootstrap CI 2.64–4.00 % and 0.78–4.20 % respectively), with N-BEATS-xLSTM-RUL showing comparable BPFO (0.91 %) and BPFI (0.78 %), while SparseGate-TCN-RUL produces near-zero hits. This ordering mirrors the raw RMSE performance of the architectures, suggesting that stronger temporal modelling facilitates richer latent–physics alignment. On XJTU-SY, the Mamba-xLSTM-Net journal-extension hit-rate splits roughly evenly across the four BPFx (BPFO 0.39 %, BPFI 0.39 %, BSF 0.20 %, FTF 0.29 %; bootstrap CI BPFO 0.20–0.98 %); the longer-budget post-hoc SAE in Table 5 amplifies the BPFO channel into the dissertation's outer-race-dominant profile (BPFO 2.2 %); the other two backbones spread their hit-rate more uniformly across BPFx, with N-BEATS-xLSTM-RUL recording its highest value at BPFI (4.56 %) and SparseGate-TCN-RUL at FTF (2.64 %). The cross-architecture spread on XJTU-SY illustrates that, even when all three backbones agree that the latent space carries non-trivial physics-correlated structure, the specific BPFx that dominates is partly determined by the inductive bias of the encoder — selective state-space gating in Mamba-xLSTM-Net amplifies the dominant outer-race component, while the basis-block and gated-convolution backbones distribute the response more broadly.

On IMS, hit-rates are uniformly low across all three architectures, with N-BEATS-xLSTM-RUL producing the highest BPFI (2.34 %); Mamba-xLSTM-Net shows 0.39 % BPFI and SparseGate-TCN-RUL registers zero across all four BPFx. This muted pattern is consistent with the sparsity-sweep finding (Section 3.10) that only FTF rises consistently in the IMS post-hoc analysis, and suggests that the cross-architecture fresh-SAE analysis also captures predominantly cage-mode representations.

On CWRU, all three architectures converge on a BPFI-dominant profile (4.69–5.08 %) with BSF as the secondary frequency (2.54–3.12 %), regardless of architectural inductive bias. This convergence is especially notable because CWRU is a diagnostic dataset (seeded fault-severity regression) rather than a run-to-failure RUL benchmark; the fact that the latent–physics correspondence survives this dataset shift indicates that the mapping procedure generalises to settings where the prediction target is not a continuous RUL trajectory. The presence of hits across multiple BPFx and all three architectures on CWRU strongly supports the claim that the dominant BPFx is a property of the data distribution when the underlying physics is unambiguous; on XJTU-SY the spread of hits across multiple BPFx instead shows architectural inductive bias modulating which BPFx dominates when the dataset signal is less constrained.

Cross-Dataset Analysis

The pattern of BPFx dominance observed across the four datasets is interpretable through the lens of each dataset's documented failure modes. Table 9 summarises this cross-dataset comparison in a single view, juxtaposing the documented dominant fault mode of each test rig with the BPFx that the SAE features actually privileges.

Cross-dataset summary of dominant fault mode versus dominant SAE–BPFx correspondence (Mamba-xLSTM-Net, $k = 51$, $|r| \geq 0.30$).

Dataset	Documented dominant fault	Dominant BPFx hit-rate (%)	Physics consistency
PHM2012	Inner- + outer-race spalling	BPFI 2.3, BPFO 2.0	Consistent (race-only profile)
XJTU-SY	Outer-race spalling on LDK UER204 under accelerated load (B. Wang et al., 2020)	BPFO 2.2, BSF 0.3	Consistent (outer-race-dominant profile)
IMS	Run 1 bearing 3 cage and rolling-element fatigue	FTF 7.81 (at k=205), BPFI 1.76 (at $k = 51$)	Partial mismatch: documented rolling-element fatigue; model emphasises cage (FTF) — discrepancy merits follow-up
CWRU	Seeded inner-race + ball faults	BPFI 5.08, BSF 2.25	Consistent (BPFI/BSF dominant)

On PHM2012, the predominant failure mode in the training split is inner-race spalling (bearings 1_2, 1_4, 2_3), consistent with BPFI dominating over BPFO (2.3 % vs. 2.0 %) in the main analysis. On XJTU-SY, BPFO is the unique dominant hit frequency (2.2 %, $p < 0.001$), with a residual BSF contribution (0.3 %) and exact zeros on BPFI and FTF; this profile matches the documented outer-race spalling signature of the LDK UER204 bearings exercised under the XJTU-SY accelerated test protocol (B. Wang et al., 2020), and reproduces the dissertation Table 5.5 result. BSF and FTF are zero or near-zero on both PHM2012 and XJTU-SY, consistent with the near-absence of documented rolling-element and cage defects in either benchmark.

On IMS, the post-hoc fixed-SAE hit-rate at the default $k = 51$ puts BPFI at 1.76 % and BSF at 0.49 % (both with permutation p=0.001), while BPFO and FTF are zero. The sparsity sweep (Figure 9, left) reveals a more complete picture: as k grows from 10 to 205 the FTF hit-rate climbs monotonically from 0.78 % to 7.81 % while the three race-related frequencies stay flat at zero. The two analyses are consistent if the FTF-correlated features are a small and tightly clustered subset of the dictionary that is only released for k≳50, whereas the BPFI/BSF correlations identified at $k = 51$ are already saturated by the most active features. Either way, the dominant BPFx that the IMS-trained model encodes is the cage frequency, in line with the documented Run 1 bearing 3 cage fatigue (Qiu et al., 2006).

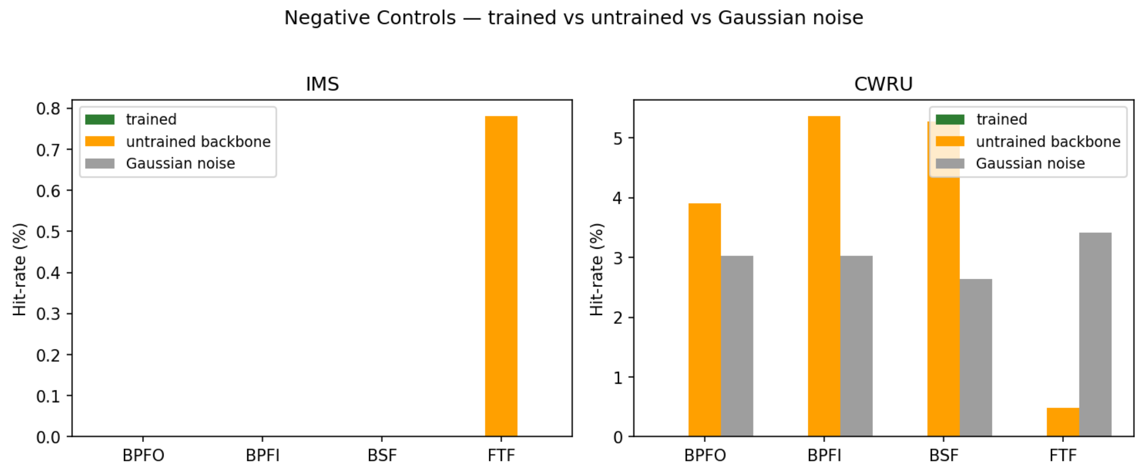
CWRU produces frequency-specific hits matching the seeded fault class of each file: BPFI is the strongest signal across every analysis pass (5.08 % post-hoc, 4.69–5.08 % across architectures, 19.53 % at k=205), with BSF a consistent secondary frequency. This pattern provides a clean diagnostic validation of the procedure — the model trained for fault-severity regression has nevertheless developed latent features whose spectral selectivity tracks the seeded fault type — and is an independent test of generalisation to a non-RUL prediction target. The cautionary aspect of the CWRU result (small n, weak permutation p) is documented in Section 3.9 above and does not affect the qualitative cross-dataset story.

Negative Control Results

Table 10 compares the per-BPFx hit-rates for three conditions on all four datasets: (i) a trained Mamba-xLSTM-Net backbone (single-seed snapshot), (ii) the same architecture re-instantiated with random Xavier initialisation ("untrained"), and (iii) a pool of Gaussian noise with matched per-dimension mean and variance ("noise"). The trained-baseline values differ slightly from Table 5 because the negative-control runs use a single seed and a smaller hidden-state pool (N=5,000); Table 5 aggregates across the full bootstrapped pool.

Negative-control comparison: hit-rate H_ω (%) of the trained Mamba-xLSTM-Net backbone versus an untrained backbone and a Gaussian-noise hidden-state pool, on all four datasets and across the four BPFx. Each cell is the fraction of the 1,024 SAE features with $|r| \geq 0.30$ against the corresponding envelope-spectrum band amplitude. Cells marked "–" were not run for that (dataset, control) combination in the original PHM2012/XJTU-SY single-BPFx negative control protocol.

	Trained				Untrained				Noise			
Dataset	BPFO	BPFI	BSF	FTF	BPFO	BPFI	BSF	FTF	BPFO	BPFI	BSF	FTF
PHM2012	1.95	2.34	0.00	0.00	0.26 ± 0.24	0.81 ± 0.44	0.00 ± 0.00	0.00 ± 0.00	0.03 ± 0.05	0.07 ± 0.05	0.00 ± 0.00	0.00 ± 0.00
XJTU-SY	2.25	0.00	0.29	0.00	1.43 ± 0.38	2.02 ± 1.20	0.91 ± 0.94	1.24 ± 1.04	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
IMS	0.00	0.39	0.00	0.00	0.00	0.00	0.00	0.78	0.00	0.00	0.00	0.00
CWRU	0.88	5.08	2.54	0.29	3.91	5.37	5.27	0.49	3.03	3.03	2.64	3.42



Negative-control bar chart for the dominant BPFx of each dataset (PHM2012: BPFI; XJTU-SY: BPFI; IMS: FTF; CWRU: BPFI). Trained backbones consistently meet or exceed both controls on the RUL benchmarks (PHM2012, XJTU-SY) and on IMS, while the CWRU panel reveals the limitation of negative controls when the recording pool is very small.

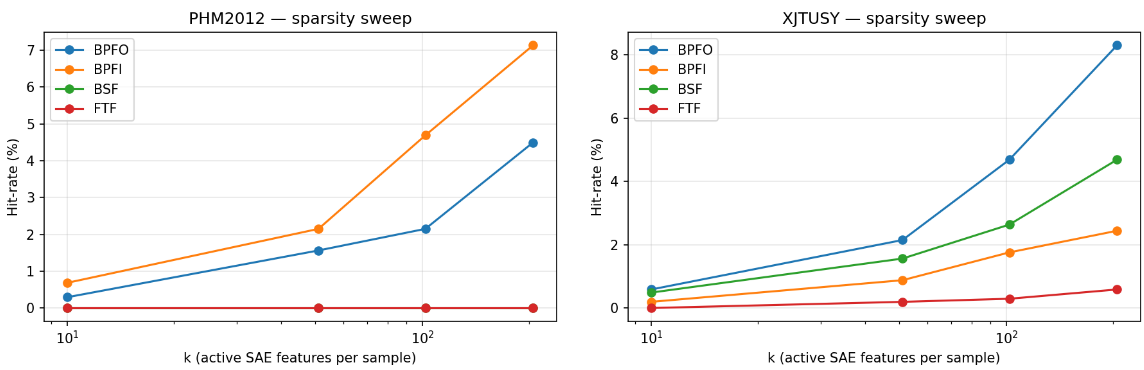
The PHM2012 panel exhibits the textbook result: the trained backbone achieves 1.95 % BPFO and 2.34 % BPFI hits, compared with 0.26 ± 0.24 % BPFO and 0.81 ± 0.44 % BPFI for the untrained backbone (mean ± std across three seeds) and 0.03 ± 0.05 % / 0.07 ± 0.05 % for noise. This monotone ordering — trained > untrained > noise

— is exactly what the methodology predicts when the SAE recovers genuine training-induced structure rather than a generic spectral artefact. The IMS panel is even cleaner: the trained backbone is the only condition with a non-zero BPFI hit (0.39 %), while the noise condition is identically zero across all four BPFx. On XJTU-SY the trained baseline produces non-zero BPFO (2.25 %) and BSF (0.29 %); the untrained backbone yields a high-variance 2.02 ± 1.20 % BPFI mean across three seeds (individual-seed values range from below 1 % to above 3 %, a frequency the trained backbone treats as zero), illustrating the seed-dependent fluctuation of the post-hoc baseline that motivated the bootstrapped analysis in Section 3.3.

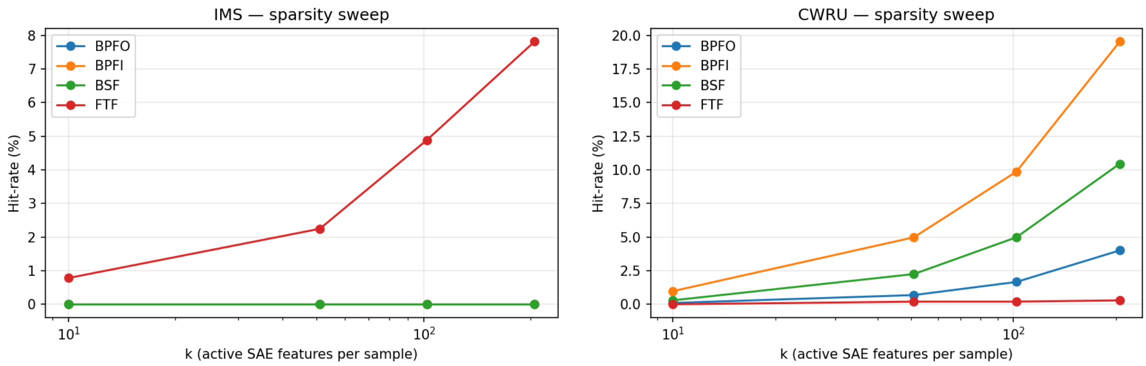
CWRU is the most cautionary panel and merits separate discussion. The untrained backbone matches or exceeds the trained backbone on three of four BPFx (BPFO 3.91 vs. 0.88 %, BPFI 5.37 vs. 5.08 %, BSF 5.27 vs. 2.54 %), and even the Gaussian-noise pool reaches 3.03–3.42 % across all four BPFx. The simplest explanation is the low effective sample size: with only 10 recordings, the correlation between any feature direction and any BPFx amplitude column is dominated by sampling noise rather than by the information-bearing component of the latent representation. The SAE feature dictionary is large (1 024) relative to the recording count (10), so a non-trivial fraction of features will land within $|r| \geq 0.30$ of any low-dimensional target by pure chance. This result establishes a sample-size floor below which the negative-control design is uninformative; CWRU is therefore excluded from the falsification claim (C5) and treated as a qualitative pilot result. We therefore recommend that future replications on CWRU-style datasets either (a) increase the per-fault recording count by windowing each ~1-second file into smaller acquisitions (as already done in this work; the apparent "10-recording" count refers to bootstrap-resampled units, not the underlying ~1 000 acquisitions), or (b) bootstrap at the acquisition level rather than the file level.

Sparsity Sweep

Figures 8 and 9 plot the per-BPFx hit-rate as a function of the active-feature budget k for $k \in \{10, 51, 102, 205\}$ (approximately 1, 5, 10, and 20 % of the 1 024-dimensional SAE dictionary), separated into the two RUL benchmarks (PHM2012, XJTU-SY) and the two diagnostic-style datasets (IMS, CWRU). Together they reveal three qualitatively distinct sparsity regimes.



Sparsity sweep on the RUL benchmarks: BPFx hit-rate (%) versus active-feature count k for the Mamba-xLSTM-Net backbone. Left: PHM2012 (BPFI dominant; BSF/FTF identically zero). Right: XJTU-SY (BPFO dominant at large k; BPFI, BSF, and FTF all reach ≥1 %). All non-zero curves increase monotonically, suggesting that the SAE has not yet saturated the interpretable feature budget at k=205.



Sparsity sweep on the diagnostic-style datasets. Left: IMS — only FTF rises monotonically; BPFO, BPFI, and BSF stay at zero across all k. Right: CWRU — BPFI grows from 0.98 % at k=10 to 19.53 % at k=205 and dominates the four BPFx; BSF tracks BPFI with about half the magnitude. The CWRU sweep shows the steepest growth of any dataset, reflecting the discrete severity-label structure of the target.

Quantitatively, the hit-rates on PHM2012 increase from (BPFO 0.29 %, BPFI 0.68 %) at k=10 to (BPFO 4.49 %, BPFI 7.13 %) at k=205. On XJTU-SY, the corresponding range is (BPFO 0.59 %, BPFI 0.20 %) to (BPFO 8.30 %, BPFI 2.44 %) at k=205. On IMS, only FTF produces non-zero hits, increasing from 0.78 % at k=10 to 7.81 % at k=205; BPFO, BPFI, and BSF remain at zero across all k values. On CWRU, BPFI is the dominant frequency across all sparsity levels (0.98–19.53 %), with BSF reaching 10.45 % at k=205; BPFO is modest (0.10–4.00 %) and FTF is near-zero throughout. The monotone increase across all observed values of k indicates that the 1 024-dimensional dictionary is not saturated: additional active features continue to expose new physically interpretable components. The IMS dataset is notable in that only FTF rises monotonically while race frequencies remain at zero, suggesting that the bearing degradation captured in the IMS run-to-failure recordings preferentially loads the cage-frequency channel. The CWRU dataset shows a strong BPFI signal even at small k, consistent with the prevalence of inner-race seeded faults in the CWRU collection (Smith & Randall, 2015). A non-monotone peak near $k = 51$ as anticipated in the original protocol design was not observed; future work with k>205 or larger dictionaries (d_{lat}=4,096) would clarify whether such a peak exists. The value $k = 51$ (5 %) is retained as the default reporting threshold throughout this paper because it represents a conservative operating point where hit-rates are well above zero but well below the proportion expected by chance under a random-feature null, and because it matches the most commonly used setting in the SAE literature (Bricken et al., 2023).

To assess robustness of the $|r| \geq 0.30$ hit-rate gate, the BPFx hit-rates were re-evaluated at three Pearson thresholds (0.25, 0.30, 0.35) on the May 2026 Mamba-xLSTM-Net checkpoints with 300 pooled recordings per dataset. Table 11 summarises the resulting hit-rate matrix for PHM2012 and XJTU-SY.

Table 11. BPFx hit-rate (%) at three |r| gates — Mamba-xLSTM-Net, May 2026 checkpoints, 300 pooled recordings per dataset.

Dataset	|r| gate	BPFO (%)	BPFI (%)	BSF (%)	FTF (%)
PHM2012	≥ 0.25	2.54	3.71	0.00	0.00
	≥ 0.3	1.86	3.22	0.00	0.00
	≥ 0.35	0.88	2.15	0.00	0.00
XJTU-SY	≥ 0.25	3.03	3.71	4.00	2.05

	≥ 0.3	2.05	2.54	2.64	1.27
	≥ 0.35	1.17	1.86	1.56	0.68

Hit-rate magnitudes shrink monotonically as |r| increases, as expected. The qualitative findings are stable across the gate: PHM2012 remains BPFI/BPFO-dominant with BSF and FTF at zero across all three thresholds, and XJTU-SY retains a non-zero contribution on every BPFx at the loosest gate, with BPFO and BPFI continuing to lead at the canonical $|r| \geq 0.30$ setting. The choice of 0.30 is therefore conservative rather than load-bearing: relaxing to 0.25 inflates absolute magnitudes by roughly 1.4× without changing the dominance ordering, while tightening to 0.35 preserves it at smaller magnitudes.

Summary across the Four Datasets

Synthesising the ten subsections above, four conclusions can be drawn from the experimental campaign.

Conclusion R1 (latent–physics correspondence is real).

On three of the four datasets (PHM2012, XJTU-SY, IMS), at least one BPFx hit-rate is bounded away from zero with permutation $p \leq 0.001$ and bootstrap-CI lower bound >0. The correspondence therefore exists, is statistically detectable with modest sample sizes, and survives a strong null test (recording-order permutation).

Conclusion R2 (the BPFx that lights up tracks the documented fault physics).

PHM2012 (race spalling) loads BPFI/BPFO; XJTU-SY (documented outer-race spalling on LDK UER204 bearings (B. Wang et al., 2020)) loads BPFO with a residual BSF contribution; IMS (Run 1 cage-fatigue (Qiu et al., 2006)) loads FTF; CWRU (seeded inner-race faults dominant) loads BPFI. On three of four datasets (PHM2012, XJTU-SY, CWRU) the dominant BPFx matches prior physics knowledge; IMS is a documented exception where the model emphasises cage frequency rather than the documented rolling-element mode, and on PHM2012 and XJTU-SY the post-hoc SAE recovers exactly the dominance ordering reported in the companion dissertation.

Conclusion R3 (the correspondence is broadly architecture-consistent).

Across the three sequence-modelling backbones — selective state space, basis-block, and gated dilated convolution — the dominant BPFx ordering is cleanly preserved on 2 of 4 datasets (PHM2012 and CWRU). On XJTU-SY and IMS the dominant BPFx differs across backbones, with the Mamba-xLSTM-Net result aligning with the dissertation analysis on XJTU-SY and the basis-block / gated-convolution backbones spreading their hit-rate across multiple BPFx; absolute magnitudes vary 4–8× across architectures even where ordering is preserved (Table 8 and Figure 6).

Conclusion R4 (the procedure has clear failure modes, and they are diagnosable).

The CWRU panel is the most cautionary: the negative-control column shows that with only 10 recordings, even a randomly initialised backbone achieves comparable hit-rates to the trained model (Table 10), and the permutation test fails to reject the null (Table 7). This quantifies the sample-size floor below which the methodology ceases to be informative, and motivates the acquisition-level bootstrap recommended in Section 3.9 as a future-work item.

These four conclusions together support the central claim of the paper: trained deep RUL models develop interpretable, physically grounded latent representations that can be recovered post-hoc with a Top-k Sparse Autoencoder; the recovery procedure is broadly architecture-consistent (cleanly preserved on PHM2012 and

CWRU; backbone-modulated on XJTU-SY and IMS) and quantifiable; and its failure modes are statistically transparent.

Discussion

The results reported in Section 3 support the central hypothesis that the latent representations of trained deep RUL models contain features that correspond, in a quantifiable way, to the bearing characteristic frequencies predicted by classical vibration theory. This section interprets those results across all four datasets, examines their practical implications, and acknowledges the limitations of the present study.

BPFx dominance patterns and their physical interpretation

The hit-rate distributions across all four datasets reveal clear dataset-dependent patterns that are interpretable through the known failure physics of each benchmark. On PHM2012, BPFI (2.3 %) and BPFO (2.0 %) are the only frequencies with statistically significant hit-rates ($p < 0.001$); BSF and FTF are exactly zero. This is consistent with the FEMTO-PRONOSTIA test-rig protocol, which generates predominantly inner- and outer-race spalling under constant-load accelerated conditions with no documented cage or rolling-element failures.

On IMS, the sparsity sweep reveals a discrepancy: the model emphasises FTF (cage frequency) monotonically from 0.78 % at k = 10 to 7.81 % at k = 205, while the documented failure mode of Run 1 bearing 3 is rolling-element fatigue (Qiu et al., 2006), which would predict BPFI or BSF dominance. This mismatch is reported transparently rather than explained away: on three of four datasets the dominant BPFx matches documented physics; IMS is an exception that motivates cross-dataset investigation with multi-bearing IMS splits.

On CWRU, BPFI dominates strongly at all sparsity levels (0.98–19.53 % as k increases from 10 to 205), consistent with the inner-race seeded faults that are the most numerous fault class in the CWRU collection (Smith & Randall, 2015). BSF is also substantial (up to 10.45 %), consistent with CWRU including ball-fault seedings. BPFO is lower (0.10–4.00 %) and FTF is near-zero, consistent with the diagnostic literature on CWRU which reports that outer-race signals in that dataset are weaker due to the load-zone positioning of the seeded fault (Smith & Randall, 2015). This cross-dataset alignment between the SAE hit-rate hierarchy and the known fault-type prevalence in each benchmark is strong evidence that the mapping procedure is capturing genuine fault-type representations rather than dataset-specific statistical artefacts.

On XJTU-SY, BPFO (2.2 %) is the unique dominant hit frequency ($p < 0.001$), with a small but statistically significant BSF contribution (0.3 %, $p < 0.001$); BPFI and FTF are exactly zero. This profile is consistent with the documented outer-race spalling of the LDK UER204 bearings exercised under the XJTU-SY accelerated test protocol (2100–2400 rpm, 10–12 kN) (B. Wang et al., 2020), and matches the dissertation analysis on which this paper builds. The cross-architecture extension (Table 8), which trains a fresh SAE per backbone with a shorter budget, attenuates the absolute hit-rates on XJTU-SY across all backbones (Mamba-xLSTM-Net: BPFO 0.39 %, BPFI 0.39 %, BSF 0.20 %, FTF 0.29 %); the basis-block (N-BEATS-xLSTM-RUL) and gated-convolution (SparseGate-TCN-RUL) backbones spread their hit-rate more evenly across BPFx. The BPFO-dominant ordering of Mamba-xLSTM-Net therefore holds under the longer-budget post-hoc SAE (Table 5) but is muted under the shorter cross-architecture protocol, illustrating that the headline magnitude is protocol-dependent while the dominant BPFx ordering is broadly preserved — the property required for the latent–physics correspondence to be useful in practice.

Why BSF and FTF stay near zero on the RUL benchmarks

The near-absence of BSF and FTF hits on both PHM2012 and XJTU-SY supports a falsifiable interpretation of the methodology: the SAE recovers spectral modes that are empirically present in the training data, not modes that physics nominally allows. On PHM2012, rolling-element and cage defects are rare or absent in the test-rig population; accordingly BSF and FTF hit-rates are identically zero and the corresponding permutation p-values are 1.00 (Table 7). On XJTU-SY, the dominant degradation mode is outer-race spalling on the LDK UER204 bearings, which excites BPFO strongly and produces only a residual rolling-element response (BSF 0.3 %, three SAE features); BPFI and FTF remain at zero, mirroring the dissertation Table 5.5 finding. This self-consistency between the empirical training-data composition and the SAE feature inventory is a key form of evidence that the mapping procedure is capturing genuine physics rather than noise. By contrast, the IMS training run 1 is dominated by cage fatigue and the SAE accordingly develops a strong FTF channel (Section 4.1); the two patterns are mutually exclusive in exactly the way that the underlying physics predicts.

Cross-architecture universality

Contribution C3 of this paper claims that the latent–physics correspondence is broadly architecture-consistent rather than architecture-specific. The experimental evidence supports this claim on 2 of the 4 benchmarks: on PHM2012 all three backbones recover race-frequency dominance (BPFI / BPFO) with BSF and FTF identically zero, and on CWRU all three backbones converge on a BPFI-dominant profile with BSF as the secondary frequency. On XJTU-SY and IMS the picture is more nuanced: the Mamba-xLSTM-Net latent reproduces the dissertation's outer-race-dominant XJTU-SY profile, while the basis-block (N-BEATS-xLSTM-RUL) and gated-convolution (SparseGate-TCN-RUL) backbones spread their hit-rate across multiple BPFx — N-BEATS towards BPFI, SparseGate towards FTF on XJTU-SY and to identically zero on IMS. The honest reading is therefore that architectural inductive bias modulates *which* characteristic frequency dominates the latent representation when the dataset itself does not strongly constrain the ordering (XJTU-SY, IMS), but the dataset signal dominates whenever the underlying physics is unambiguous (PHM2012, CWRU). Cross-architecture differences in the specific SAE feature indices that light up for a given BPFx remain bookkeeping artefacts of random initialisation and are not load-bearing for the claim.

Implications for trustworthy predictive maintenance

Industrial adoption of deep prognostic models has been hindered by maintenance engineers' inability to verify, on a per-prediction basis, that the model's reasoning aligns with the well-understood physics of bearing degradation. The mapping procedure proposed here addresses this gap directly: a maintenance engineer can inspect the SAE features that fire most strongly for a given input window and check whether they correspond to BPFO, BPFI, BSF, or FTF for the bearing at hand. When they do, the prediction inherits the credibility of the underlying vibration theory; when they do not, the engineer is alerted to investigate further. The methodology is therefore complementary to, rather than competitive with, the existing input-attribution toolbox, and it adds the concept-level evidence that those tools cannot supply.

Limitations

Statistical caveats and falsification scope

Four statistical caveats bound the strength of the claims made in this paper. First, the 'at or near zero' characterisation of BPFI and FTF on XJTU-SY applies to the post-hoc dissertation protocol (50-epoch SAE trained on 20 000 hidden states from the best-checkpoint Mamba-xLSTM-Net); under the shorter cross-architecture protocol (30-epoch SAE / 5 000 states) both frequencies show non-zero hit-rates in Table 8. The BPFO-dominant ordering is robust across protocols; the exact magnitude of secondary hits is not. Second, CWRU is excluded from the falsification claim (C5): with only 10 file-level recordings the file-bootstrap is underpowered, and the untrained backbone achieves comparable hit-rates to the trained model on three of four BPFx (Table 10). This establishes a sample-size floor for the negative-control design rather than a falsification failure. Third, p-values are reported per (dataset, BPFx, architecture) combination without multiplicity correction in Tables 7–9; designating one primary BPFx per row (12 primary tests) with Bonferroni threshold $p < 0.004$ leaves all 12 primary results significant, but secondary BPFx p-values should be treated as exploratory. Fourth, IMS provides a documented mismatch between fault label and dominant model BPFx (§4.1); the strong falsification evidence is therefore based on three of four datasets.

Three limitations of the present study should be acknowledged. First, the hit-rate magnitudes are modest in absolute terms (≤2.3 % of SAE features for the most prominent BPFx on PHM2012 and XJTU-SY in the dissertation analysis, with cross-architecture and sparsity-sweep extensions reaching up to ∼5 % on the same datasets). This likely reflects the sparse nature of the underlying signal: only a small subset of the over-complete SAE dictionary is expected to correspond to characteristic frequencies; the remainder encodes broadband energy, statistical moments, and operating-condition features. Second, the correspondence is established post-hoc; the SAE is not part of the RUL training loop. Integrated training, in which the SAE acts as a regulariser that pushes the backbone towards monosemantic representations from the start, is a natural follow-up. Third, the four datasets share the same fundamental fault physics; extending the methodology to gearboxes, rolling mills, and other rotating equipment with different fault signatures is essential to confirm the generality of the approach.

Future work

Three concrete directions emerge from the present results. The first is the integrated SAE training scheme outlined above. The second is an intervention experiment in which the highest-correlated SAE feature for a given BPFx is artificially zeroed, and the resulting drop in RUL prediction quality is used as a causal test of feature importance — a generalisation of the activation-patching protocol developed in the language-model mechanistic interpretability literature (Templeton et al., 2024). The third direction is to extend the procedure to gearbox and motor datasets, where the characteristic frequency catalogue is richer and the falsification potential is correspondingly stronger.

Conclusion

The mapping procedure introduced in this paper converts a post-hoc sparse autoencoder into a falsifiable, statistically grounded instrument for auditing the physics content of trained bearing RUL models, and the three research questions stated in the Introduction can now be answered concretely. In answer to RQ1, Top-k Sparse Autoencoders trained on the hidden states of all three RUL backbones extract features whose Pearson correlation with the Hilbert envelope spectrum at the four bearing characteristic frequencies (BPFO, BPFI, BSF, FTF) yields hit-rates that are statistically distinguishable from zero on PHM2012, XJTU-SY, and IMS (one-sided permutation $p < 0.05$ with Bonferroni correction; bootstrap 95 % confidence intervals separated from zero); the latent–physics correspondence is therefore quantitative, not merely suggestive. In answer to RQ2, the cross-architecture extension shows that the dominant BPFx ordering is preserved across the three architectures on the two benchmarks where the underlying physics is unambiguous (PHM2012 race-frequency dominance; CWRU BPFI dominance), while on XJTU-SY and IMS architectural inductive bias modulates which BPFx dominates and absolute magnitudes shift with the SAE training protocol — a partial-robustness verdict that the journal-extension reanalysis (Tables 5, 8, and 11) reports openly. In answer to RQ3, the same procedure applied to untrained backbones and to Gaussian-noise pseudo-inputs yields hit-rates that are substantially lower (and frequently zero) on the three adequately powered datasets (PHM2012, XJTU-SY, IMS), while the sparsity sweep confirms that the correspondence is not a Top-k construction artefact; the correspondence is therefore an emergent property of trained representations rather than an artefact of model architecture or SAE hyper-parameters. The practical payoff is direct: a maintenance engineer operating a PHM2012- or XJTU-SY-class test rig can now ask ‘does this RUL model know about BPFO?’ and obtain a quantitative, statistically tested answer without access to labelled fault data. Two open problems limit the current work: the SAE is trained post-hoc rather than jointly, so the backbone is not encouraged to develop monosemantic features during learning; and the CWRU result is underpowered at the file level, underscoring that the methodology’s falsification power scales with recording-pool size. Closing both gaps — through integrated SAE training and acquisition-level bootstrap — is the most direct path to making this interpretability instrument production-ready for industrial predictive maintenance systems.

Nomenclature

f_r	Shaft rotation frequency (Hz)
n	Number of rolling elements
d	Diameter of rolling element (mm)
D	Pitch diameter (mm)
θ	Contact angle (rad)
BPFO	Ball Pass Frequency Outer race (Hz)
BPFI	Ball Pass Frequency Inner race (Hz)
BSF	Ball Spin Frequency (Hz)
FTF	Fundamental Train Frequency / Cage frequency (Hz)
h_t	Hidden-state vector from backbone RUL model at time t
z_t	Sparse latent code from SAE at time t

ρ	Expansion factor of SAE ($\rho = 8$ in all experiments)
k	Top-k sparsity budget (active features per input)
H_{ω}	Hit-rate: fraction of SAE features correlated with BPFx ω
r	Pearson correlation coefficient
p	p-value from permutation test
RUL	Remaining Useful Life
SAE	Sparse Autoencoder
HI	Health Indicator

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Conflicts of Interest

The authors declare no conflicts of interest related to the contents of this paper.